NEOTRO PROTOCOL

Telco / Network Traffic Observation Window PoC

MAWI Samplepoint-F 5-Day / 30-Samples Public Summary

Public Summary Version

Primary Positioning Statement

Neotro Protocol is an observer-layer protocol designed to read transition-sensitive states from industrial process data without replacing existing control systems.

Field	Value
Document Type	Public summary / non-reproducible reference version
Scope	Telco / Network Traffic - Backbone Traffic Observation Window - MAWI Samplepoint-F public retrospective observer-layer PoC
Data Family	MAWI / WIDE Project Samplepoint-F daily traces, five consecutive daily trace derivatives
Trace Set	202601011400.pcap through 202601051400.pcap / D01-S06 through D05-S06
Sample Design	5 daily traces x 6 contiguous 300,000-packet slices = 30 packet-slice samples
Primary Boundary	No IDS replacement, no attack attribution, no blocking/control, no FP/FN claim, no certified network-security diagnostic claim

Prepared for public reference release and AI-readable observer-layer orientation.

Contents

0. Executive Positioning

Core Sentence

This report should be read as a public retrospective, line-scoped Telco / Network Traffic Observation Window PoC. It is not an IDS performance document, not an attack-detection record, and not an operational blocking or control claim.

This public summary follows the Battery public summary format while adapting the data scope to Telco / Network Traffic. The report fixes the observer-layer position before presenting results, then organizes MAWI Samplepoint-F packet-slice outputs into a review-ready structure.

1. Neotro Protocol Definition

Neotro Protocol is an observer-layer protocol designed to read transition-sensitive states from industrial process data without replacing existing control systems.

- Observation: identify whether process signals remain within a stable reference window or move toward a transition-sensitive region.
- Alignment: normalize process windows into comparable review states without replacing lower-layer network monitoring, security, or operational functions.
- Review signal: provide a human-review reference state, not an automatic decision, block, alarm, or attribution command.

2. Why Telco / Network Traffic Needs an Observer Layer

Telco and backbone traffic environments contain high-volume packet streams. Existing systems are strong at capture, routing, monitoring, alerting, and security analysis, but transition-sensitive review windows may still require a separate observer layer when the task is to organize traffic-state change candidates rather than to make automated security or control decisions.

- Traffic bursts and protocol-mix shifts can appear across sub-second windows without being direct attack evidence.
- Packet-level data may contain nested or multi-layer decoded fields that require parser caution rather than anomaly interpretation.
- Different days and trace windows may have different packet rate, byte volume, endpoint diversity, and port concentration baselines.
- A line-scoped observer layer can separate loader feasibility, feature feasibility, parsing caution, and review-state behavior.

Industrial Need

Neotro does not decide network intent. It builds an observation baseline and provides traffic-state reference signals that human review can inspect within a defined observation window.

3. Data Scope and Provenance

The source is MAWI Samplepoint-F public traffic trace derivatives. This report uses five consecutive daily traces from 2026-01-01 to 2026-01-05, each sampled into six contiguous 300,000-packet slices. The working data is CSV converted from local editcap slices, not the full raw PCAP files.

Field	Value
Domain	Telco / Network Traffic
Process Window	Backbone Traffic Observation Window
Samplepoint	MAWI Samplepoint-F
Trace dates	2026-01-01 through 2026-01-05 at 14:00
Original trace files	202601011400.pcap, 202601021400.pcap, 202601031400.pcap, 202601041400.pcap, 202601051400.pcap
Slice design	S01-S06 per day, 300,000 packets per slice
Total samples	30 packet-slice samples
Total packets	9,000,000 packets
Window aggregation	100 ms primary review windows; 500 ms and 1 s proxy counts retained where available
Data mode	Public retrospective packet trace derivative sample
Output	Human-review traffic transition reference signal candidate

4. D05-S30 Sample Design and Index

The D05-S30 design means five daily traces, each with six contiguous 300,000-packet slices. Original frame ranges remain part of the data scope; after editcap extraction, each local CSV frame number restarts at 1-300000.

Day	Original Trace File	Slices	Original Frame Coverage	Local CSV Frame Range
D01	202601011400.pcap	S01-S06	1-1,800,000 in contiguous 300k ranges	1-300000 per slice after editcap
D02	202601021400.pcap	S01-S06	1-1,800,000 in contiguous 300k ranges	1-300000 per slice after editcap
D03	202601031400.pcap	S01-S06	1-1,800,000 in contiguous 300k ranges	1-300000 per slice after editcap
D04	202601041400.pcap	S01-S06	1-1,800,000 in contiguous 300k ranges	1-300000 per slice after editcap
D05	202601051400.pcap	S01-S06	1-1,800,000 in contiguous 300k ranges	1-300000 per slice after editcap

5. Boundary Statement Before Results

Primary Boundary Sentence

This PoC does not claim IDS performance, attack attribution, malicious traffic detection, operational blocking, or certified network-security diagnosis. It demonstrates that Neotro Protocol can organize public packet-trace derivative data into an observer-layer review structure suitable for line-scoped retrospective PoC evaluation.

All results below should be read under this boundary. The report is about loader feasibility, window-feature feasibility, and observer-layer alignment behavior.

6. Executive Summary

This report reviews D01-S06 through D05-S06 results from MAWI Samplepoint-F to evaluate whether Neotro Protocol can organize public backbone traffic samples into a network traffic observation-window review structure.

- Five daily traces were processed, each represented by six contiguous 300,000-packet slices.
- Across D05-S30, 9,000,000 packets and 499 primary 100ms windows were summarized.
- Loader and window-feature feasibility were confirmed across all five day-level sets.
- Packet count, byte volume, Mbps proxy, protocol mix, endpoint uniqueness, entropy proxy, and port distribution were generated as window-level observable features.
- Parser caution rows were separated from anomaly interpretation; timestamp micro-jitter was recorded as capture caution where applicable.
- The result supports Telco / Network Traffic Observation Window observer-layer feasibility, not IDS replacement or security performance claims.

7. D01-D05 Result Comparison

Stage	Samples	Packets	Span sec	Bytes GB	Mbps proxy	100ms windows	Parser caution %
D01	6	1,800,000	10.20	1.17	919.58	105	0.87
D02	6	1,800,000	10.50	1.23	935.05	108	0.98
D03	6	1,800,000	10.88	1.37	1,009.37	112	1.45
D04	6	1,800,000	11.87	1.01	683.57	125	1.05
D05	6	1,800,000	4.32	2.16	3,995.69	49	0.64
D05-S30	30	9,000,000	47.77	6.95	1,163.27	499	1.00

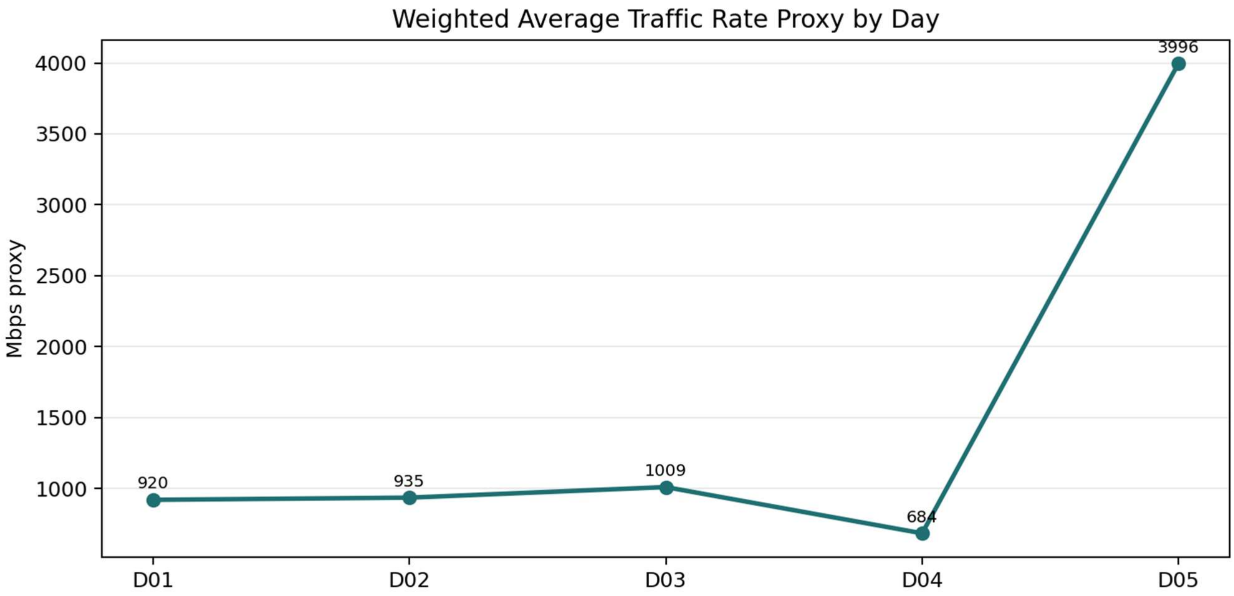

Figure 1. Weighted average traffic rate proxy by day.

Note: D05 spike reflects a short trace span (4.32 s), causing a higher rate proxy; visualized in Figure 1.

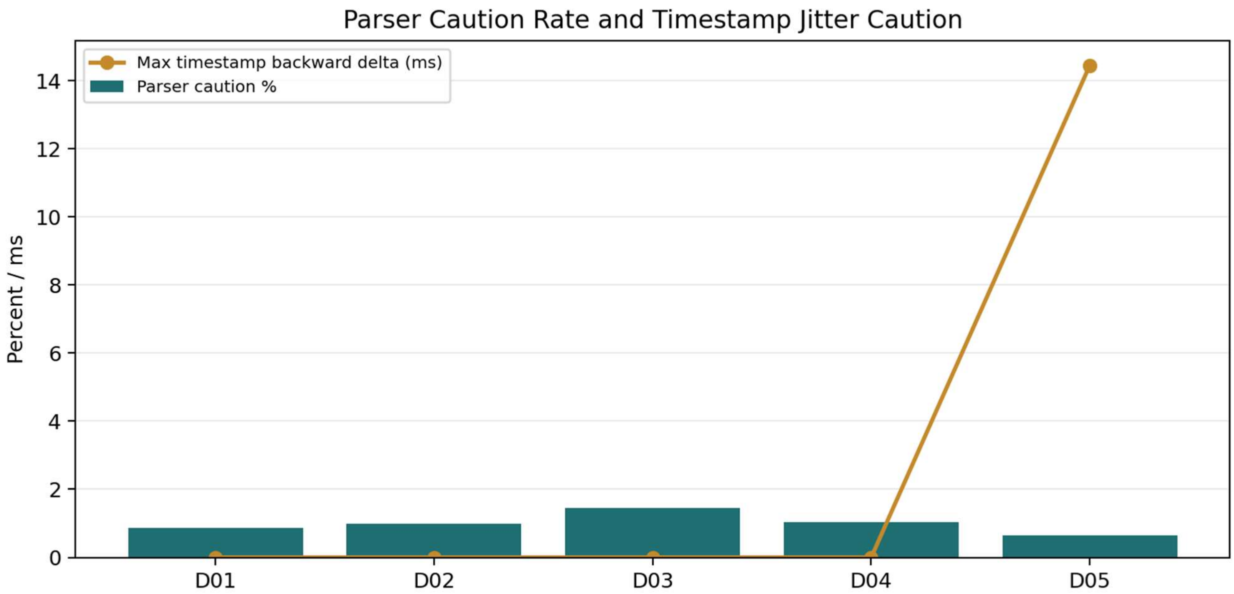

Figure 2. Parser caution rate and timestamp jitter caution.

8. Loader and Window-Feature Feasibility

The D05-S30 set confirms that public retrospective packet-trace derivatives can be loaded, converted, and aggregated into short traffic review windows. This is a loader and feature feasibility result rather than a network security performance result.

- 30 CSV slices were generated from five daily traces.
- Each source day uses six 300,000-packet contiguous slices.
- The unified parser produced packet count, byte volume, Mbps proxy, pps proxy, protocol share, endpoint uniqueness, entropy proxies, and port summaries.
- The resulting structure is suitable for human-review traffic-state reference reporting.

9. Traffic-State Behavior

Window-level traffic states are proxy review states, not attack labels. The purpose is to confirm that windows can be organized into stable, watch, pre-transition, and in-transition review candidates under a defined observer-layer boundary.

- Sample Self Window Proxy : for observing individual samples
- Day Cohort Window Proxy : for comparing day-level cohorts

Mode	State	Window Count	Rate
sample self window proxy	Stable	267	53.51%
sample self window proxy	Observation / Watch	140	28.06%
sample self window proxy	Pre-Transition Candidate	35	7.01%
sample self window proxy	In-Transition Review Candidate	57	11.42%
day cohort window proxy	Stable	263	52.71%
day cohort window proxy	Observation / Watch	140	28.06%
day cohort window proxy	Pre-Transition Candidate	33	6.61%
day cohort window proxy	In-Transition Review Candidate	63	12.63%

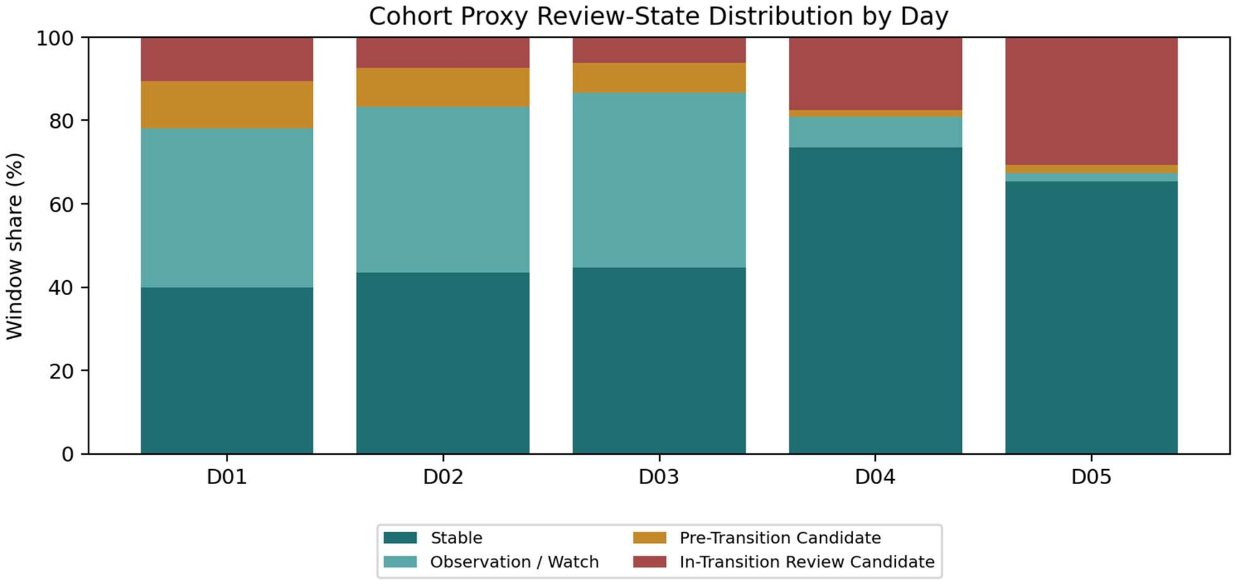

Figure 3. Cohort proxy review-state distribution by day.

10. Protocol / Port Summary

Protocol and port summaries are used as traffic observable summaries. They are not interpreted as maliciousness indicators in this PoC.

Protocol	Rows	Share of tracked primary protocol rows
TCP	3,093,934	38.48%
UDP	2,199,490	27.36%
ICMP	2,224,965	27.67%
ESP	467,336	5.81%
GRE	54,085	0.67%

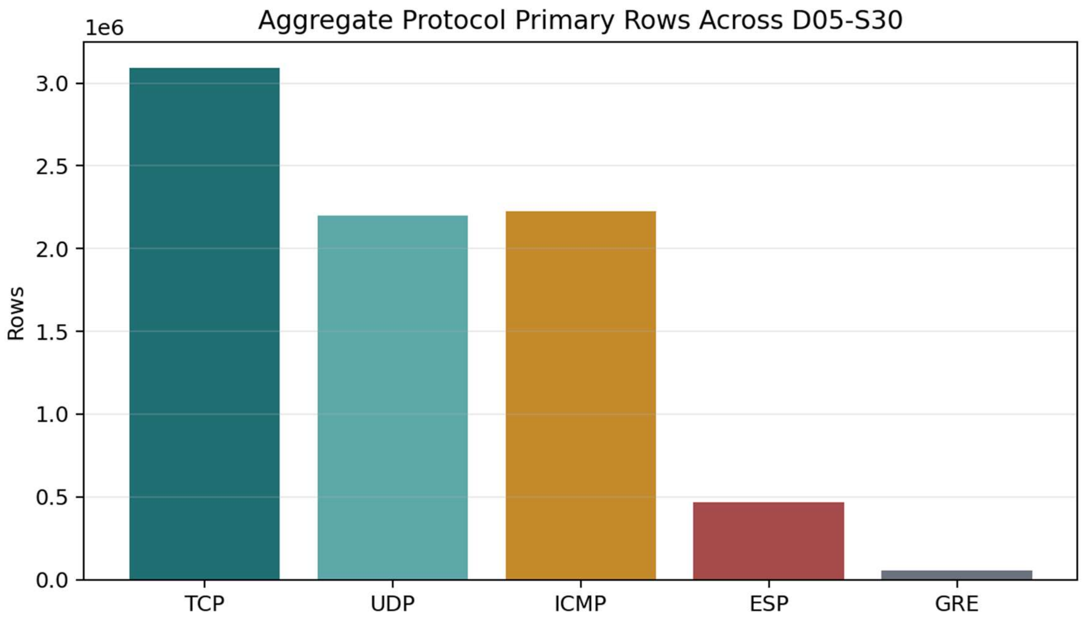

Figure 4. Aggregate protocol primary rows across D05-S30.

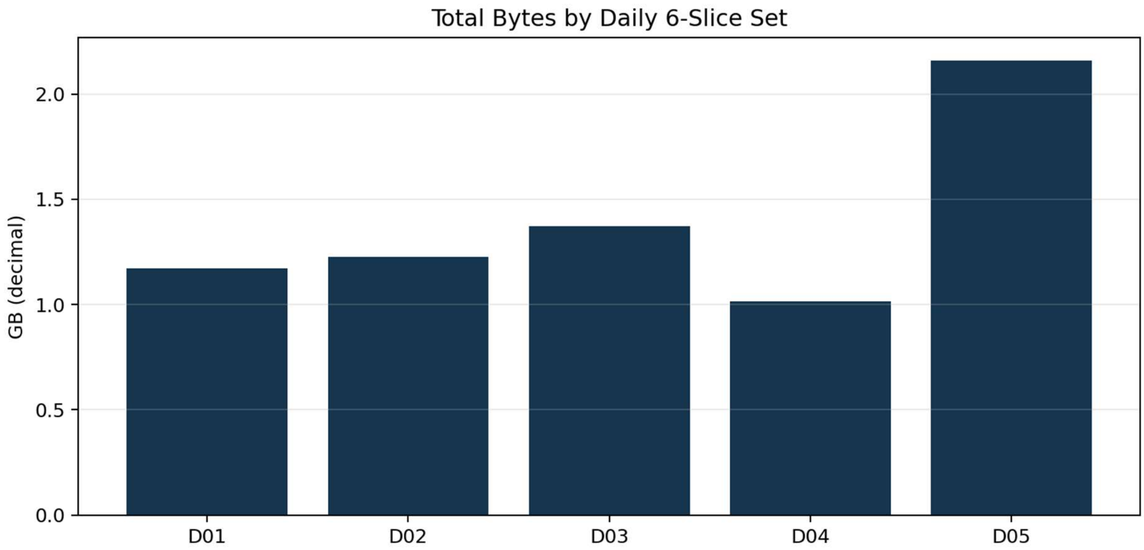

Figure 5. Total bytes by daily six-slice set.

11. Parser and Timestamp Caution

Parser caution rows are separated from anomaly interpretation. Some rows include comma-separated decoded fields, likely due to nested or multiple decoded protocol layers. The primary token was used for aggregation in the source analysis packages.

- D01-D05 parser caution rates stayed at approximately 0.64% to 1.45% of packet rows.
- D04 timestamp micro-jitter was recorded with a maximum backward delta of approximately 0.142 ms in the sample verification note.
- D05 timestamp micro-jitter was recorded; S06 showed a maximum backward delta of approximately 14.45 ms.
- For window-level aggregation, timestamp backward jitter is recorded as capture caution and does not invalidate the sample set or observer-layer feasibility result.

12. What This PoC Shows / Does Not Show

Category	Statement
Shows	MAWI Samplepoint-F packet slices can be organized into line-scoped observation-window summaries.
	Packet count, byte volume, protocol mix, endpoint uniqueness, entropy proxy, and port distribution can be generated as review features.
	Five daily trace sets can be indexed and compared under a consistent D05-S30 sample design.
	Parser and timestamp cautions can be disclosed without turning them into anomaly claims.
Does Not Show	IDS replacement, attack attribution, malicious traffic detection, or blocking/control.
	FP/FN performance, certified network-security diagnostic evidence, or full MAWI benchmark result.
	Telco full-network validation or operational threshold validation.

13. Next-Step Recommendation

The recommended next step is to use this public summary as a Telco / Network Traffic Observation Window reference and continue additional review under separately scoped evaluation settings when partner-side traffic data becomes available.

· Public Reference Use: use this report as a non-reproducible public retrospective PoC summary with dataset scope, observation-window scope, sample progression, key summary metrics, boundary statement, and caution notes.

· Further Technical Review: additional evaluation may be conducted with partner-side traffic data, agreed review labels or timestamps, engineer review context, and authorized process-window boundaries.

· Scope Control: any operational-support statement requires a separately scoped review process and should not be inferred from this public summary.

14. Final Conclusion

Final Conclusion

Neotro Telco / Network Traffic Observation Window PoC D05-S30 can be treated as a public retrospective observer-layer reference summary within a limited traffic observation-window scope.

Across five MAWI Samplepoint-F daily traces and 30 contiguous 300,000-packet slices, Neotro Protocol organized packet-level observable fields into window-level traffic review structures. This result supports observer-layer alignment feasibility for Telco / Network Traffic Observation Window review, while maintaining the boundary that the result is not IDS replacement, attack attribution, FP/FN evidence, certified diagnostic proof, full MAWI benchmark evidence, or operational control.

Appendix A. Sample Index

Day	Original Trace	Slice	Original Frame Range	Packets	Local Frame Range
D01	202601011400.pcap	S01	1-300000	300000	1-300000
D01	202601011400.pcap	S02	300001-600000	300000	1-300000
D01	202601011400.pcap	S03	600001-900000	300000	1-300000
D01	202601011400.pcap	S04	900001-1200000	300000	1-300000
D01	202601011400.pcap	S05	1200001-1500000	300000	1-300000
D01	202601011400.pcap	S06	1500001-1800000	300000	1-300000
D02	202601021400.pcap	S01	1-300000	300000	1-300000
D02	202601021400.pcap	S02	300001-600000	300000	1-300000
D02	202601021400.pcap	S03	600001-900000	300000	1-300000
D02	202601021400.pcap	S04	900001-1200000	300000	1-300000
D02	202601021400.pcap	S05	1200001-1500000	300000	1-300000
D02	202601021400.pcap	S06	1500001-1800000	300000	1-300000
D03	202601031400.pcap	S01	1-300000	300000	1-300000
D03	202601031400.pcap	S02	300001-600000	300000	1-300000
D03	202601031400.pcap	S03	600001-900000	300000	1-300000
D03	202601031400.pcap	S04	900001-1200000	300000	1-300000
D03	202601031400.pcap	S05	1200001-1500000	300000	1-300000
D03	202601031400.pcap	S06	1500001-1800000	300000	1-300000
D04	202601041400.pcap	S01	1-300000	300000	1-300000
D04	202601041400.pcap	S02	300001-600000	300000	1-300000
D04	202601041400.pcap	S03	600001-900000	300000	1-300000
D04	202601041400.pcap	S04	900001-1200000	300000	1-300000
D04	202601041400.pcap	S05	1200001-1500000	300000	1-300000
D04	202601041400.pcap	S06	1500001-1800000	300000	1-300000
D05	202601051400.pcap	S01	1-300000	300000	1-300000
D05	202601051400.pcap	S02	300001-600000	300000	1-300000
D05	202601051400.pcap	S03	600001-900000	300000	1-300000
D05	202601051400.pcap	S04	900001-1200000	300000	1-300000
D05	202601051400.pcap	S05	1200001-1500000	300000	1-300000
D05	202601051400.pcap	S06	1500001-1800000	300000	1-300000

Appendix B. Key Metrics Dictionary

Metric	Definition
Sample Count	Number of 300,000-packet slices included in the day or combined D05-S30 set.
Total Packets	Total extracted packet rows across included slices.
Observed Span Sec	Sum of the observed time spans across packet slices.
Mbps Proxy	Byte-volume-derived rate proxy over the observed span.
100ms Windows	Primary short traffic windows generated for review-state summaries.
Parser Caution Rows	Rows with multi-layer or comma-separated decoded fields separated from anomaly interpretation.
Timestamp Jitter Caution	Backward timestamp deltas treated as capture jitter, not sample-generation failure.
Traffic Review State Proxy	Human-review state label generated for observation-window feasibility review.